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CSIT 598 Weekly Quiz

Multiple Choice Questions

1. Which statement best characterizes a Convolutional Neural Network (CNN)?
   (a) A model that learns temporal hierarchies by stacking recurrent cells only.
   (b) A model that learns spatial hierarchies of features through layers of filters (convolutions), often for image data.
   (c) A linear model equivalent to logistic regression with hand-crafted features.
   (d) A tree-based ensemble for classification and regression.

2. For a 2D input of size 7×7, using a single 3×3 filter, stride $= 2$, and no padding, what is the spatial size of the resulting feature map?

   (a) 5×5
   (b) 4×4
   (c) 3×3
   (d) 2×2

3. In standard image CNNs, how do convolutional filters relate to the input depth (channels)?
   (a) Each filter covers exactly one input channel only.
   (b) Filters are 1×1 by definition, so depth is irrelevant.
   (c) Each filter extends across the full input depth (e.g., all channels).
   (d) Depth sharing means filters randomly skip channels during training.

4. What is the primary purpose of max pooling in CNNs?
   (a) To increase the spatial resolution of feature maps.
   (b) To reduce spatial dimensions and emphasize dominant activations per feature map.
   (c) To normalize activations to zero mean and unit variance.
   (d) To connect every neuron to the entire input volume.

Written Question

Convolutional Neural Networks

Consider a convolutional neural network with the following architecture:

1. Input layer: a 7×7 input map:

$$X = \begin{bmatrix} 1 & 2 & 3 & 0 & 1 & 2 & 3 \\ 4 & 5 & 6 & 1 & 2 & 3 & 4 \\ 7 & 8 & 9 & 2 & 3 & 4 & 5 \\ 0 & 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 4 & 5 & 6 & 7 & 8 & 9 \end{bmatrix}$$

(1)

$$X \xrightarrow{F_1} X_1 = \begin{pmatrix} -6 & 12 & -6 \\ -6 & 2 & -6 \\ -6 & -6 & -6 \end{pmatrix} \xrightarrow{\text{ReLU}} X_2 = \begin{pmatrix} 0 & 12 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

2. Convolutional layer:
   - One 3×3 filter with stride $= 2$ and no padding

$$F_1 = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}$$

   - ReLU activation applied after convolution
3. Pooling layer: max pooling with a 2×2 filter and stride $= 1$
4. Fully connected layer:
   - Flatten the output of pooling layer to a vector
   - The weights are $[-0.1 \;\; 0.3 \;\; 0.2 \;\; -0.2]$

Solve the following questions:

1. Perform the convolution operation using the given filter F_1. Then apply ReLU activation. What is the resulting feature map?
2. Apply 2×2 max pooling with stride 1 to the feature maps. What is the pooled feature map?
3. Flatten the pooled feature map into a single vector. Compute the output of the fully connected layer.

(2) $A_2 \xrightarrow{\text{MaxPooling}} \begin{bmatrix} 12 & 12 \\ 2 & 2 \end{bmatrix}$

(3) - $\hat{y} = 12 \times 0.1 + 12 \times 0.3 + 0.2 \times 2 - 0.2 \times 2 = 2.4.$